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### Final Project Proposal

For my final project, I want to make an algorithm that helps the user organize their bookshelf by their preference of height, genre, author, alphabetical order or not, series vs standalone, and ascending vs descending. While most people do not have personal libraries in their homes, those who do could sometimes be overwhelmed by how to organize their shelves. I personally have this issue because I have over 300 books with limited space, and I always get overwhelmed on how to fit all of my books on my bookshelves while trying to keep a certain aesthetic to them. Readers really care about their books and the appearance of their bookshelves, so I want to make an algorithm that will help me and hopefully others if this turns out to be a good product.

My idea on this algorithm is to allow the user to input the books they want to organize and the bookshelf length and height they want to fit their books on. Then they will be allowed to make a hierarchical order on how they want their shelves to be organized. For example, I organize my bookshelves by first genre, then by series, and finally height. Other people could prefer first author, then height, and finally series, whereas others could only care about the author. From there, the algorithm will then provide an order of the books on the shelf based on the user's preference—I am still deciding if I should provide a visual of this order to the user.

After doing some surface level research, I have come to the conclusion that I will need to use data structures, like strings for the name of the books, author, genre, and series, integers for

the number of books, their order in a series, and the height of the books, and enum to use for categorizing between standalone vs series. The time complexity of this algorithm will depend on how many preferences the user has. The more categories the user has, the worse the time complexity will be, I believe. I know that I need to compare elements, like the author of books, so I need to make a comparison function using the dictionary of a book and the preference the user wants to organize their shelves.

To ensure the correctness of this algorithm, AI has told me that Python has a built-in library called *unittest* that will test if the code works as expected. I will need to make sure that my logic is correct so that the *unittest* will work. I will have a dummy test and see if it will organize it correctly. In regards to measuring performance, I will use *timeit*.